[GAME NAME HERE]: An Educational Bubble-shooter game centered around Changes in Matter

A Capstone Project

Presented to

The Department of Software Technology
College of Computer Studies
De La Salle University

In partial fulfillment
Of the requirements for the degree of

Bachelor of Science in Interactive Entertainment
Major in Game Development

UBAY, Conrad Amadeus Carmelo C.

DIMAUNAHAN, Ryan
Adviser

December 5, 2025

ABSTRACT

Chemistry education contains foundational concepts that are often perceived as difficult for students to understand, particularly in terms of their application to real-world situations. This difficulty may lead to cognitive overload, especially when instruction lacks opportunities for practical engagement. Interactive learning approaches allow students to actively apply and develop their analytical skills while discovering how chemistry concepts relate to everyday experiences.

This study explores the use of Game-Based Learning (GBL) through the development of [GAME NAME HERE], an educational serious game designed as a learning material for Grade 6 students under the MATATAG curriculum in the Philippines. The game focuses on physical and chemical changes of matter, expanding students' understanding beyond the three common states of matter. Through interactive gameplay, students engage with concepts such as phase changes and heat transfer while learning to distinguish physical changes from chemical changes.

The primary goal of this study is to improve students' conceptual understanding, motivation, and emotional engagement by reinforcing hands-on learning and real-world applications of chemistry concepts. This project aims to provide both students and teachers with an accessible and engaging instructional resource that supports visualization, experimentation, and the development of analytical skills in basic chemistry education.

Keywords: physical and chemical changes, game-based learning, educational serious game
Sustainable Development Goals: 4: Quality Education, 11: Sustainable Cities and Communities, 12: Responsible Consumption and Production

ABSTRAK

Ang pag-aaral ng kimika ay naglalaman ng mga pundamental na konsepto na kadalasang itinuturing na mahirap unawain ng mga mag-aaral, lalo na pagdating sa aplikasyon nito sa totoong buhay. Ang ganitong kahirapan ay maaaring magdulot ng cognitive overload, partikular kung kulang ang mga aralin sa praktikal at interaktibong gawain. Sa pamamagitan ng interaktibong pagkatuto, aktibong nahahasa ng mga mag-aaral ang kanilang kasanayan sa pagsusuri at mas nauunawaan nila kung paano nagagamit ang mga konsepto ng kimika sa pang-araw-araw na buhay.

Sinisiyasat ng pag-aaral na ito ang paggamit ng Game-Based Learning (GBL) sa pamamagitan ng pagbuo ng [GAME NAME HERE], isang pang-edukasyong seryosong laro na idinisenyo bilang pantulong na materyal sa pagkatuto para sa mga mag-aaral sa Baitang 6 sa ilalim ng MATATAG kurikulum sa Pilipinas. Nakatuon ang laro sa pisikal at kemikal na pagbabago ng materyal, na nagpapalawak sa kaalaman ng mga mag-aaral lampas sa tatlong karaniwang kaanyuan ng matter. Sa pamamagitan ng interaktibong gameplay, natututuhan ng mga mag-aaral ang mga konsepto tulad ng pagbabago ng estado at paglipat ng init, habang malinaw na natutukoy ang kaibahan ng pisikal at kemikal na pagbabago.

Layunin ng pag-aaral na ito na mapabuti ang pag-unawa, motibasyon, at emosyonal na pakikipag-ugnayan ng mga mag-aaral sa pamamagitan ng hands-on na pagkatuto at aplikasyon ng mga konsepto ng kimika sa mga sitwasyong hango sa totoong buhay. Nilalayon din ng proyektong ito na magsilbing kapaki-pakinabang na materyal para sa mga guro at mag-aaral upang mas mapadali ang pagkatuto ng kimika sa isang biswal, interaktibo, at makabuluhang paraan.

Mga susing salita: pisikal at kemikal na pagbabago, game-based learning, pang-edukasyon na laro

Contents

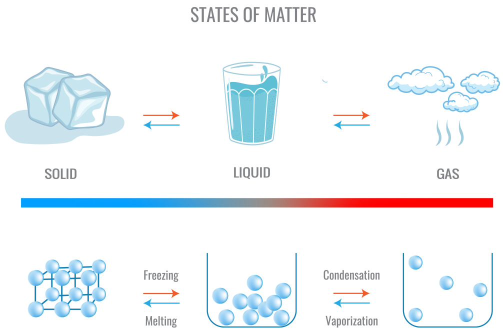
List of Figures

List of Tables

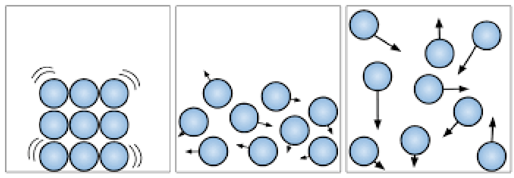
CHAPTER 1
INTRODUCTION

1.1 Background of the Study

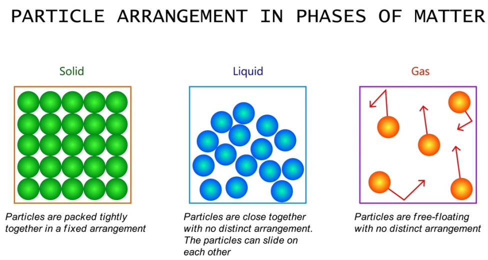
Science education at the elementary level serves as a foundational stage in developing scientific literacy, conceptual reasoning, and problem-solving skills. Within the MATATAG Curriculum of the Department of Education (DepEd), Grade 6 Science introduces learners to fundamental concepts related to materials, including the description of changes of state for solids, liquids, and gases—specifically melting, evaporation, freezing, and condensation—using diagrams and flowcharts. These competencies aim to help learners understand observable phenomena through scientific representations and structured reasoning.



(a) Phase change diagram



(b) Phase changes (Khan Academy)



(c) Phase changes (88Guru)

Figure 1.1: Examples of phase change diagrams used in science education

Despite the importance of these competencies, research has consistently shown that students experience difficulty in understanding the particulate nature of matter and the mechanisms underlying phase changes. Learners often memorize definitions of melting or evaporation without fully comprehending the particle-level interactions that explain these transformations. Misconceptions frequently arise regarding particle spacing, motion, and the role of heat transfer in altering states of matter (Adadan, 2013; Paik, Kim, Cho, & Park, 2004). These challenges are compounded by the abstract nature of microscopic representations, which are not directly observable.

In many classroom settings, demonstrations of phase change are limited to water-based experiments, such as melting ice or boiling water. While these activities provide observable macroscopic changes, they often lack explicit integration with particle-level models, resulting in fragmented conceptual understanding. Furthermore, access to laboratory materials for demonstrating processes such as sublimation or deposition is limited at the elementary level, restricting exposure to diverse examples of state change.

Game-Based Learning (GBL) has emerged as a promising instructional approach that integrates interactive systems with curricular objectives. Educational games can provide dynamic visualizations, immediate feedback, and experiential learning opportunities that reduce cognitive overload and promote conceptual change (Plass, Homer, & Kinzer, 2015). When designed with explicit learning outcomes and aligned assessments, digital games may enhance engagement and deepen understanding, particularly in abstract scientific domains.

This study introduces [GAME NAME HERE], a 2D educational platformer developed as a supplementary and enrichment learning tool aligned with the MATATAG Grade 6 Science competency on changes of state. The game integrates explicit terminology (e.g., "Solid $\rightarrow$ Liquid: Melting") while allowing learners to discover appropriate phase changes through environmental problem-solving. Central to its design is an interactive particle-lattice minigame in which learners manipulate bonds and particle spacing to simulate state transitions. By directly engaging with representations similar to those used in classroom diagrams, learners repeatedly encounter visual models of particle behavior in a contextualized, goal-driven environment.

Given the increasing emphasis on learner-centered instruction under the MATATAG framework, there is a need to explore innovative digital tools that complement formal instruction without replacing it. This study examines whether a platformer-based educational game can support conceptual understanding, engagement, and reinforcement of curriculum-aligned competencies related to phase changes of matter.

1.2 Statement of the Problem

This study seeks to investigate the potential of a platformer-based educational game as a supplementary instructional tool for teaching changes of state in Grade 6 Science.

Specifically, the study aims to answer the following questions:

1. To what extent does the use of [GAME NAME HERE] improve learners' conceptual understanding of changes of state (melting, evaporation, freezing, and condensation) at the particle level?

2. How does interaction with the particle-lattice minigame influence students' ability to relate macroscopic phase changes to microscopic particle behavior?

3. What are the perceptions of students and parents regarding engagement, usability, and perceived learning value of the game?

4. How feasible is the implementation of a discovery-based yet explicitly labeled digital game in a supplementary learning context?

1.3 Objectives of the Study

The general objective of this study is to design and evaluate a curriculum-aligned educational platformer that reinforces Grade 6 competencies on changes of state.

Specifically, the study aims to:

1. Develop [GAME NAME HERE] as a supplementary digital learning tool aligned with the MATATAG Grade 6 Science competency on describing changes of state using diagrams and flowcharts.
2. Integrate an interactive particle-level simulation that visually represents solids, liquids, and gases and their transitions.
3. Assess learners' conceptual understanding before and after gameplay using structured instruments.
4. Measure student engagement and perceived motivation through survey responses.
5. Collect qualitative insights from students and parents regarding usability and learning reinforcement.

1.4 Scope and Delimitations

This study focuses exclusively on the development and evaluation of [GAME NAME HERE], a 2D platformer designed to reinforce the Grade 6 Science competency on changes of state of matter. The game addresses physical changes only and does not include instruction on chemical reactions or chemical change.

The participant pool may include Grade 6 learners as well as students from adjacent grade levels who have previously completed instruction on changes of state. Sampling will follow a convenience-based recruitment model due to logistical constraints. The study adopts a quasi-experimental single-group design incorporating pre- and post-assessments, engagement surveys, and qualitative interviews with students and parents.

The study does not measure long-term retention or academic performance beyond the duration of the intervention. Additionally, the research evaluates the game as a supplementary enrichment tool and does not compare it directly with traditional instruction through a control group.

1.5 Significance of the Study

This study contributes to elementary science education by examining the integration of interactive digital modeling within a platformer game environment. For learners, the game provides repeated exposure to particle representations within meaningful problem-solving contexts, potentially strengthening conceptual connections between diagrams and observable phenomena.

For educators, the study offers insights into how educational games may complement MATATAG-aligned instruction without replacing formal teaching. The findings may inform future curriculum integration strategies and digital resource development.

For researchers, the study adds to the growing body of literature on game-based learning in science education, particularly within the Philippine educational context. By examining discovery-based mechanics combined with explicit labeling, the study contributes to ongoing discussions on balancing constructivist learning with guided instruction in digital environments.

1.6 Definition of Terms

Changes of State – Physical transformations between solid, liquid, and gas phases, including melting, evaporation, freezing, and condensation.

Particle-Level Representation – Visual models depicting spacing, motion, and bonding of particles to represent different states of matter.

Game-Based Learning – An instructional approach that integrates educational objectives within interactive digital game environments.

Supplementary Learning Tool – An instructional resource designed to reinforce and enrich formal classroom instruction without replacing it.

CHAPTER 2
REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Review of Related Literature

2.1.1 Conceptual Change in Science Education

Learners frequently enter science classrooms with pre-existing misconceptions about natural phenomena. In the context of phase changes, students often misunderstand particle behavior, energy transfer, and state transitions at the molecular level. Conceptual Change Theory posits that meaningful learning occurs when learners confront inconsistencies in their existing mental models and reconstruct their understanding based on intelligible and plausible alternatives.

Interactive simulations have been shown to support conceptual restructuring by making abstract microscopic processes visible. By allowing learners to observe and manipulate particle arrangements during phase transitions, digital environments can facilitate deeper conceptual understanding and correction of misconceptions.

This theoretical perspective informs the educational foundation of the present study.

2.1.2 Constructivism and Interactive Learning

Constructivist Learning Theory asserts that knowledge is actively constructed through interaction and experience rather than passively received. Game-based environments embody constructivist principles by placing learners in active problem-solving roles.

In platform-based game environments, players experiment, test strategies, and receive immediate feedback from system responses. This aligns with the constructivist view that learning occurs through engagement and reflection upon action.

The interactive particle manipulation mechanics of the developed game reflect this constructivist orientation.

2.1.3 Multimedia Learning in Science Education

Multimedia Learning Theory suggests that learning is enhanced when verbal and visual information are presented in coordinated and complementary ways. Effective multimedia design integrates text, visuals, and animation to support meaningful cognitive processing.

The game integrates explicit textual labeling (e.g., "Solid $\rightarrow$ Liquid: Melting") alongside dynamic molecular visualization. This dual-channel presentation supports the formation of coherent mental representations of microscopic processes.

2.1.4 Game-Based Learning and Engagement

Game-Based Learning Theory emphasizes the role of challenge, feedback, and goal-oriented interaction in promoting learner engagement. Educational games combine instructional objectives with gameplay mechanics to foster sustained attention and intrinsic motivation.

Engagement is a critical factor in educational technology adoption. Games that balance challenge and usability are more likely to maintain player interest and promote exploratory interaction.

This study evaluates engagement using structured post-play instruments to determine whether the finalized build demonstrates positive user experience.

2.1.5 Experiential Learning and Simulation

Experiential Learning Theory posits that learning occurs through cycles of experience, reflection, conceptualization, and experimentation. Simulated environments allow learners to experience phenomena that may not be directly observable in traditional classrooms.

By simulating molecular interactions during freezing, melting, and evaporation, the game provides experiential exposure to abstract scientific processes.

2.1.6 Outcome-Based Education, Bloom's Taxonomy, and MATATAG Curriculum

Outcome-Based Teaching and Learning emphasizes alignment between instructional activities and clearly defined competencies. The MATATAG Curriculum for Grade 6 Science outlines competencies related to understanding states of matter and phase transitions.

Bloom's Taxonomy further categorizes cognitive processes from remembering to applying and analyzing. The game's level design encourages players to apply conceptual knowledge to solve in-game environmental challenges, aligning with higher-order cognitive objectives.

2.1.7 Iterative Design and User-Centered Game Development

In game development research, iterative design is a widely accepted approach for refining gameplay mechanics, usability, and user experience. Iterative development involves repeated cycles of testing, evaluation, modification, and regression validation.

Campaign testing refers to structured playthrough sessions aimed at identifying usability issues, progression bottlenecks, and gameplay friction points. Regression testing ensures that modifications introduced in later builds do not reintroduce previously resolved issues.

User-centered design principles emphasize incorporating player feedback into successive iterations to improve overall game quality.

The present study adopts an iterative development framework prior to conducting a final structured evaluation of the game's usability and engagement.

2.2 Review of Related Studies

2.2.1 International Studies

International research has demonstrated the effectiveness of educational games in enhancing engagement and conceptual understanding in science subjects. Studies evaluating interactive simulations and digital learning games report improvements in learner motivation and positive user experience when multimedia elements and active manipulation are integrated.

Additionally, research on usability evaluation models such as MEEGA+KIDS highlights the importance of assessing engagement, focused attention, satisfaction, and perceived learning in educational game contexts.

Studies on iterative game development emphasize that structured playtesting contributes significantly to improved playability, stability, and user satisfaction.

2.2.2 Local Studies

Local studies in educational technology within the Philippine context indicate increasing integration of digital learning tools aligned with curriculum standards. Research evaluating instructional games commonly employs usability surveys and engagement instruments to assess learner reception.

However, limited research combines structured iterative development practices with formal engagement evaluation within educational game development projects.

This gap supports the relevance of the present study.

2.3 Review of Related Studies

2.3.1 International Studies

International research has demonstrated the effectiveness of educational games in enhancing engagement and conceptual understanding in science subjects. Studies evaluating interactive simulations and digital learning games report improvements in learner motivation and positive user experience when multimedia elements and active manipulation are integrated.

Additionally, research on usability evaluation models such as MEEGA+KIDS highlights the importance of assessing engagement, focused attention, satisfaction, and perceived learning in educational game contexts.

Studies on iterative game development emphasize that structured playtesting contributes significantly to improved playability, stability, and user satisfaction.

2.3.2 Local Studies

Local studies in educational technology within the Philippine context indicate increasing integration of digital learning tools aligned with curriculum standards. Research evaluating instructional games commonly employs usability surveys and engagement instruments to assess learner reception.

However, limited research combines structured iterative development practices with formal engagement evaluation within educational game development projects.

This gap supports the relevance of the present study.

2.4 Theoretical Framework

This study is grounded in Conceptual Change Theory, Constructivism, and Multimedia Learning Theory as primary anchors explaining how interactive simulation may support scientific understanding.

Supporting theories include Game-Based Learning Theory and Experiential Learning Theory, which justify the use of challenge-based interactive mechanics to foster engagement.

Outcome-Based Teaching and Learning principles and Bloom's Taxonomy guide the alignment of game mechanics with Grade 6 MATATAG competencies.

In addition, iterative design principles and user-centered development frameworks support the methodological choice of conducting campaign and regression testing prior to formal usability evaluation.

Together, these frameworks explain both:

1. The pedagogical rationale of the game, and
2. The developmental methodology employed in refining the intervention.

2.5 Conceptual Framework

The study adopts an expanded Input–Process–Output (IPO) model integrating both pedagogical foundations and iterative development practices.

Figure 2.1: Conceptual Framework of the Study

2.5.1 Explanation of the Framework

The **Input** component consists of theoretical foundations (Conceptual Change, Constructivism, Multimedia Learning, Game-Based Learning, Experiential Learning), curricular alignment (MATATAG Curriculum and Bloom's Taxonomy), and user-centered design principles.

The **Process** component is divided into two phases:

1. **Iterative Development Phase**
    - Campaign testing
    - Regression testing
    - Playtesting feedback integration
    - Gameplay refinement
2. **Final Evaluation Phase**
    - Structured play session using finalized build
    - MEEGA+KIDS usability and engagement assessment
    - Qualitative feedback collection

The **Output** component includes:

- Improved gameplay stability
- Refined user experience
- Measured engagement levels
- Evaluated usability of the finalized educational platformer

The framework illustrates that pedagogical theory informs game design, iterative development refines implementation, and structured evaluation measures user engagement and usability.

CHAPTER 3
RESEARCH METHODOLOGY

3.1 Research Design

This study employed a developmental experimental research design integrating iterative game development practices with structured usability and engagement evaluation. The research was conducted in two major phases: (1) Iterative Development and Testing, and (2) Final Structured Evaluation.

The first phase focused on refining the educational platformer through multiple campaign and regression testing cycles. The second phase involved evaluating the finalized build using standardized usability and engagement instruments.

This design was selected to align with user-centered game development methodologies while maintaining academic rigor in assessing the educational game's usability and engagement potential.

3.2 Development Framework

The development of the educational platformer followed an iterative design approach grounded in user-centered development principles.

3.2.1 Iterative Development Cycles

The game underwent multiple structured development cycles consisting of:

1. **Build Implementation** – Addition or refinement of gameplay mechanics.
2. **Campaign Testing** – Structured playthrough sessions to identify usability issues, progression bottlenecks, gameplay friction, and feature clarity.
3. **Regression Testing** – Verification that newly implemented changes did not reintroduce previously resolved issues.
4. **Feedback Integration** – Refinement of mechanics, level design, UI clarity, and performance stability.

Each iteration was documented using testing logs detailing:

- Identified bugs
- Gameplay balancing concerns
- Player confusion points

- Feature stability issues
- Suggested improvements

Data from this phase were analyzed descriptively to track improvement trends across builds.

3.3 Participants

Participants were selected using convenience sampling. Approximately 20–30 individuals were recruited through personal networks, academic peers, and referrals.

Participants were not restricted to Grade 6 students due to timeline and accessibility constraints. However, respondents were considered representative users capable of evaluating usability, engagement, and clarity of the educational mechanics.

Where participants were minors, parental consent was secured prior to participation.

Different participants may have participated across iterative testing cycles. However, the final structured evaluation phase was conducted using a stable release candidate build.

3.4 Research Instruments

The study utilized both developmental testing tools and structured evaluation instruments.

3.4.1 Campaign and Regression Testing Logs

Structured testing logs were used to document:

- Bug occurrence frequency
- Severity classification (minor, moderate, critical)
- Gameplay progression issues
- UI clarity concerns
- Player navigation difficulties
- Performance stability

These logs enabled systematic tracking of improvements across development cycles.

3.4.2 MEEGA+KIDS Usability and Engagement Instrument

The finalized version of the game was evaluated using the MEEGA+KIDS model. This instrument measures:

- Fun

- Focused attention
- Perceived learning
- Usability
- Challenge
- Satisfaction

Participants responded using a Likert-scale format.

3.4.3 Qualitative Feedback Questionnaire

Participants provided open-ended responses regarding:

- Strengths of the game
- Areas requiring improvement
- Suggestions for refinement
- Perceived clarity of phase change concepts

3.5 Data Collection Procedure

The research was conducted in two phases:

3.5.1 Phase 1: Iterative Development Testing

1. A development build was deployed.
2. Participants conducted structured play sessions.
3. Campaign testing logs documented usability and gameplay issues.
4. Regression testing verified stability after modifications.
5. Revisions were implemented before the next cycle.

This phase continued until a stable release candidate was achieved.

3.5.2 Phase 2: Final Structured Evaluation

1. Participants were briefed on the study purpose.
2. Consent was obtained.
3. Participants were given 20–30 minutes to interact with the finalized build.
4. Upon completion, participants completed:
   - MEEGA+KIDS instrument
   - Qualitative feedback questionnaire
5. Responses were collected via Google Forms.

3.6 Data Analysis

Quantitative data from the MEEGA+KIDS instrument were tabulated and analyzed using Google Sheets.

The following descriptive statistics were computed:

- Mean
- Standard deviation
- Frequency distribution
- Percentage scores

Results were interpreted to determine overall engagement and usability levels of the finalized build.

Data from campaign and regression testing logs were analyzed descriptively to identify:

- Reduction in bug frequency across iterations
- Improvement in gameplay stability
- Decrease in recurring usability issues

Qualitative responses were grouped into thematic categories to identify common patterns in player feedback.

No inferential statistical comparisons between builds were conducted, as iterative development phases involved varying participant groups.

3.7 Ethical Considerations

Participation in the study was voluntary. Participants were informed of the study's purpose and their right to withdraw at any time without penalty.

No personally identifiable information was included in the analysis. For participants under the age of 18, parental consent was secured prior to participation.

All collected data were used solely for academic research purposes.

CHAPTER 4
DESIGN AND IMPLEMENTATION

CHAPTER 5
CONCLUSION AND DISCUSSION

CHAPTER 6
RECOMMENDATIONS

REFERENCES

DataTab Team. (2025). Repeated measures anova. https://datatab.net/tutorial/anova-with-repeated-measures

Hecker, J., & Kalpokas, N. (2025, February). Thematic analysis in mixed methods approach. https://atlasti.com/guides/thematic-analysis/thematic-analysis-mixed-methods